# Nicola Novello

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nicolanovello.github.io  Nicola Novello

## Research Interests

Machine unlearning, diffusion models, differential privacy, federated learning, AI safety.

## Education

**University of Klagenfurt** — *Oct. 2022 – Present*
*PhD in Information and Communication Engineering*
Supervisors: Prof. Andrea M. Tonello, Prof. Deniz Gündüz

**University of Padova** — *Oct. 2020 – July 2022*
*Master's Degree in Control Systems Engineering. Final Mark: 110/110 with honors*
Supervisor: Prof. Alessandro Chiuso

**University of Padova** — *Oct. 2017 – July 2020*
*Bachelor's Degree in Information Engineering. Final Mark: 110/110 with honors*
Supervisor: Prof. Maria E. Valcher

## Work Experience

**University of Klagenfurt** — *Klagenfurt, Austria*
*Teaching Assistant* — *Mar. 2023 – Present*
Taught the courses "Machine learning for information and communication engineering" and "Communication systems engineering"

**University of Klagenfurt** — *Klagenfurt, Austria*
*Project Assistant* — *Apr. 2025 - Dec. 2025*
Operated research for project "Power Line Communication for Metering - Data Analytics". In collaboration with Kelag International

**Princeton University** — *Princeton (NJ), US*
*Visiting Researcher* — *Jun. 2025 - Jul. 2025*
Performed research focused on neural channel capacity estimation. Started collaboration on differentially-private federated unlearning. Under the supervision of Prof. H. Vincent Poor, department of electrical engineering

**University of Klagenfurt** — *Klagenfurt, Austria*
*Project Assistant* — *Aug. 2022 - Feb. 2024*
Conducted administrative duties for the European project ERASMUS+ NEON (Network of competence on IoT)

**Trumpf** — *Vicenza, Italy*
*Student Intern* — *Mar. 2022 - June 2022*
Developed deep learning algorithm in PyTorch for bending machine trajectory adjustment. Integrated the deep learning algorithm with existing C# infrastructure

## Selected Publications

1. **Nicola Novello**, Federico Fontana, Luigi Cinque, Deniz Gundiz, and Andrea M. Tonello, "A Unified Framework for Diffusion Model Unlearning with f-Divergence", ICML 2026
2. **Nicola Novello** and Andrea M. Tonello, "$f$-Divergence Based Classification: Beyond the Use of Cross-Entropy", ICML 2024
3. **Nicola Novello**, Andrea M. Tonello, Nunzio A. Letizia, Alex Dytso, H. Vincent Poor, "Channel Capacity Estimation with Cooperative Neural Architectures: The Role of Objective Functions and f-Divergences", to appear in IEEE Transactions on Communications

4. **Nicola Novello** and Andrea M. Tonello, "Empty SPACE: Cross-Attention Sparsity for Concept Erasure in Diffusion Models", arXiv preprint arXiv:2605.10198

5. **Nicola Novello** and Andrea M. Tonello, "Robust Classification with Noisy Labels Based on Posterior Maximization", arXiv preprint arXiv:2504.06805

6. Nunzio A. Letizia, **Nicola Novello**, and Andrea M. Tonello, "Mutual Information Estimation via $f$-Divergence and Data Derangements", NeurIPS 2024

7. Nunzio A. Letizia, **Nicola Novello**, and Andrea M. Tonello, "Copula Density Neural Estimation", IEEE Transactions on Neural Networks and Deep Learning 2025

## Other Publications

1. **Nicola Novello** and Andrea M. Tonello, "Data-Driven Grid Monitoring Under Label Noise Using Power Line Modems", IEEE SmartGridComm 2025

2. **Nicola Novello** and Andrea M. Tonello, "Label Noise Resistant $f$-Divergence Receiver for Power Line Communications", IEEE SmartGridComm 2024

3. **Nicola Novello** and Andrea M. Tonello, "Recurrent DQN for radio fingerprinting with constrained measurements collection", ICT Express (2025)

4. Frank E. Ebong, **Nicola Novello**, and Andrea M. Tonello, "Human Detection Based on Learning and Classification of Radio Scattering Parameters and Para-Hermitian Eigenvalue Decomposition", IEEE PIMRC 2024

5. **Nicola Novello** and Andrea M. Tonello, "Reinforcement learning for simplified training in fingerprinting radio localization", IEEE BalkanCom 2023

## Interests and Competences

**Programming and software skills:** Python, Matlab, C++, Java, LaTeX, GitHub
**Expertise:** Machine learning, signal processing, information theory
**Languages:** Italian (native), English (fluent)

## Academic Service

Awarded "Gold Reviewer" at ICML 2026.
Reviewer for ICML, ICLR, NeurIPS, AAAI, IEEE Transactions on Communications, IEEE Transactions on Neural Networks and Learning Systems, IEEE VTC